

Symmetry rigidity for dual characters of flagged Weyl modules, and why the cylindric M-convexity theorem is not a corollary of Fink–Mészáros–St. Dizier

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Abstract

Fink–Mészáros–St. Dizier (FMS, arXiv:1706.04935, Adv. Math. 332 (2018) 465–475) prove that for *any* diagram $D \subseteq [n]^2$ the dual character χ_D of the flagged Weyl module M_D has saturated Newton polytope, with $\text{Newton}(\chi_D) = \sum_j P(\text{SM}_n(D_j))$ a Minkowski sum of Schubert matroid polytopes. Since my cylindric M-convexity theorem (2026-09-20-c1-cylindric-M-convexity.tex, Thm. 7.1) computes exactly a saturated Newton polytope for a family of symmetric functions, the question is whether that theorem is a 2018 corollary.

The answer is no, and the reason is sharp. We prove a *symmetry rigidity* theorem: if $\text{supp}(\chi_D)$ is stable under the symmetric group permuting the coordinates it actually occupies, then every nonempty column of D is bottom-justified, and χ_D is (the support of) a straight-shape Schur polynomial $s_{\hat{\lambda}}$ whose $\hat{\lambda}$ is read off the column sizes of D . Two consequences: (i) FMS Theorem 7, restricted to symmetric dual characters, is exactly Rado’s 1952 theorem on the permutahedron, so it has no symmetric content beyond the classical; and (ii) any diagram D with $\text{supp}(\chi_D) = \text{supp}(s_{\hat{\lambda}/\mu}^c)$ *already encodes* $\hat{\lambda}$ — exhibiting D and computing $\hat{\lambda}$ are the same act — so FMS can return $\hat{\lambda}$ only if it is fed $\hat{\lambda}$.

Verification: the transcription of FMS’s definitions is calibrated against 850 divided-difference computations of Schubert and key polynomials; symmetry rigidity is confirmed exhaustively on 113,621 diagrams with no failure in either direction; the natural column dictionary fails already at $d = 0$ on 320 of 486 ordinary skew shapes, with minimal witness $\lambda = (2, 1)$, $\mu = \emptyset$, $\ell = 2$; and on 199 genuinely winding cylindric instances an exhaustive search over all diagrams finds the $\hat{\lambda}$ -skyline and nothing else. Two of the obstructions anticipated in the session brief are shown to be *false*.

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1 The question, and the form of the answer

1.1 What is being compared

My cylindric M-convexity paper proves:

Theorem 1.1 (2026-09-20-c1-cylindric-M-convexity.tex, thm:main). *Let λ/μ be a cylindric skew shape in $\mathcal{C}^{n,m}$, let $\ell \geq 1$, and suppose $\mathcal{T}_\ell(\lambda/\mu) \neq \emptyset$. Let $\hat{\lambda} = \hat{\lambda}(\lambda/\mu, \ell)$ be the sorted weight of the greedy chain. Then*

$$\text{supp}(s_{\lambda/\mu}^c(x_1, \dots, x_\ell)) = \{\alpha \in \mathbb{N}^\ell : \text{sort}(\alpha) \trianglelefteq \hat{\lambda}\} = \mathcal{P}_{\hat{\lambda}} \cap \mathbb{Z}^\ell,$$

this set is M-convex, $s_{\lambda/\mu}^c$ is SNP, and $\text{Newton}(s_{\lambda/\mu}^c) = \mathcal{P}_{\hat{\lambda}}$.

FMS prove (quoted verbatim in §2 with my own line numbers):

Theorem 1.2 (FMS Theorem 7). *Let $D = (D_1, \dots, D_n)$ be a diagram. Then χ_D has SNP, and $\text{Newton}(\chi_D) = \sum_{j=1}^n P(\text{SM}_n(D_j))$. In particular $\text{Newton}(\chi_D)$ is a generalized permutahedron.*

Theorem 1.2 holds for an arbitrary diagram, with no hypothesis whatsoever. Both theorems produce a saturated Newton polytope. The novelty question is therefore live and it is not rhetorical: if a cylindric skew shape's columns are a diagram in FMS's sense, FMS hands over the polytope and Theorem 1.1 is a 2018 corollary.

1.2 The form of the answer

The two theorems are not comparable by asking “are these the same statement?”. They are comparable by asking a single precise question:

(Q) For which diagrams D is $\text{supp}(\chi_D) = \text{supp}(s_{\lambda/\mu}^c(x_1, \dots, x_\ell))$?

If no D exists, FMS cannot reach the cylindric case. If some D exists but its construction requires $\hat{\lambda}$, FMS can be applied but returns nothing that was not put in. If some D exists and is built from $(\lambda, \mu, n, m, \ell)$ alone, Theorem 1.1 is a corollary and the registry node must be demoted on novelty.

The answer is the middle one, and §4 makes it a theorem rather than a report.

2 FMS at first hand

The source is `projects/sources/1706.04935/forArxiv.tex`, 328 lines, read in full by me on 2026-09-25. (Previously this paper stood at `agent-summary` in my index: it had been read twice by agents and never by me. The e-print v1 and v2 are byte-identical, md5 57d9ce2422cde370cc1f21a139442756, so line numbers are unambiguous.) All quotations below carry my own line numbers.

2.1 Identifier audit

The preamble (1.14–19) declares `\newtheorem{theorem}{Theorem}` with `definition`, `proposition`, `lemma`, `conjecture` and `corollary` all sharing that counter, and there is no `\numberwithin`. Counting theorem-like environments in source order:

#	line	environment	content
1	95	definition	Rothe diagram
2	146	definition	skyline diagram
3	220	definition	flagged Weyl module M_D
4	239	theorem [Kraskiewicz–Pragacz]	$\mathfrak{S}_w = \text{char}^* M_{D(w)}$
5	243	theorem [keypolynomials]	$\kappa_\alpha = \text{char}^* M_{D(\alpha)}$
6	248	definition	$\chi_D := \text{char}^* M_D$
7	253	theorem	<code>thm:Newtonofdualcharacter</code>
8	277	<i>corollary</i>	<code>thm:Schubert SNP</code>
9	293	definition [Monical–Tokcan–Yong]	the Schubitope $\mathcal{S}_D$
10	298	theorem	<code>thm:schubi</code>

So “FMS Theorem 7” is `thm:Newtonofdualcharacter`, confirmed by my own count; item 8 is a `corollary` environment, so “FMS Theorem 8” is wrong twice.

2.2 The definitions, verbatim

- **Diagram** (1.94): “By a *diagram*, we mean a subset $D \subseteq [n]^2$, a collection of boxes in the $n \times n$ grid. Throughout this paper, we view D as an ordered list of subsets $D = (D_1, D_2, \dots, D_n)$ where for each j , $D_j = \{i : (i, j) \in D\}$ is the set of row indices of boxes of D in column j .”
- **The order** $\leq$ (1.212): “For $R, S \subseteq [n]$, we say $R \leq S$ if $\#R = \#S$ and the k th least element of R does not exceed the k th least element of S for each k . For any diagrams $C = (C_1, \dots, C_n)$ and $D = (D_1, \dots, D_n)$, we say $C \leq D$ if $C_j \leq D_j$ for all $j \in [n]$.”
- **Schubert matroid** (1.193): “Fix positive integers $1 \leq s_1 < \dots < s_r \leq n$. The sets $\{a_1, \dots, a_r\}$ with $a_1 < \dots < a_r$ such that $a_1 \leq s_1, \dots, a_r \leq s_r$ are the bases of a matroid, called the *Schubert matroid* $SM_n(s_1, \dots, s_r)$.”

- **Flagged Weyl module** (1.221–223) and **dual character** (1.248–251): $M_D = \text{Span}_{\mathbb{C}}\{\prod_j \det(Y_{D_j}^{C_j}) : C \leq D\}$ and $\chi_D = \text{char}^* M_D$, where Y is the upper-triangular matrix of indeterminates and Y_R^S is the submatrix on rows S , columns R .

2.3 The one fact this paper uses

The proof of FMS Theorem 7 (1.260–275) establishes, by linear algebra and before any polytope appears:

Lemma 2.1 (FMS, 1.263–269). $\text{supp}(\chi_D) = \{\xi^C : C \leq D\}$, where $\xi_i^C = \#\{j : i \in C_j\}$; and for each j the sets S with $S \leq D_j$ are exactly the bases of $\text{SM}_n(D_j)$. Hence

$$\text{supp}(\chi_D) = \sum_{j=1}^n \{\zeta^B : B \text{ a basis of } \text{SM}_n(D_j)\} \quad (\text{sumset in } \mathbb{N}^n),$$

ζ^B the 0/1 indicator vector of B .

Everything in this paper is a statement about the right-hand side. Note in particular that $\text{supp}(\chi_D)$ is a *sumset over columns*: distinct columns contribute independently. This is the structural feature that will do all the work.

2.4 Two defects in the source, recorded

1. **A typo in the skyline formula (1.147).** FMS print

$$D(\alpha) = (D(\alpha)_1, \dots, D(\alpha)_n) \quad \text{with} \quad D(\alpha)_j = \{j \leq n : \alpha_j \geq j\}.$$

The bound variable is wrong: the intended formula is $D(\alpha)_j = \{i \leq n : \alpha_i \geq j\}$. This is forced by their own preceding sentence (1.147, “*containing the first α_i boxes in row i for each $i \in [n]$* ”) and by their own Figure 2 (1.168), whose caption reads “*The skyline diagram of $\alpha = (3, 2, 1, 0, 1)$ is $(\{1, 2, 3, 5\}, \{1, 2\}, \{1\}, \emptyset, \emptyset)$* ” — the printed formula would give $\{1\}$ for the first column. The corrected reading is what I implement, and it is independently validated in §7.1 by 698 divided-difference computations.

2. **The bibliography of FMS is defective.** `\bibitem{demazure}` reads “M. Demazure, *Key polynomials and a flagged Littlewood–Richardson rule*, Bull. Sci. Math. 98:163–172, 1974” and `\bibitem{keypolynomials}` reads “M. Demazure, *Une nouvelle formule des caracteres*, JCTA 70(1):107–143, 1995”. The titles are swapped and the second is also misattributed. The true pairing is Demazure, *Une nouvelle formule des caractères*, Bull. Sci. Math. 98 (1974) 163–172, and Reiner–Shimozono, *Key polynomials and a flagged Littlewood–Richardson rule*, JCTA 70 (1995) 107–143. Since `\cite{keypolynomials}` is the key cited for FMS Theorem 5, anyone chasing “the Demazure theorem” through this bibliography lands on the wrong paper. **FMS Theorem 5 is due to Reiner–Shimozono.**

Neither defect affects any mathematical statement of FMS. Both are recorded because I am about to cite the paper in a novelty verdict, and an identifier is not a quotation.

3 The dictionary: what each side is a function of

Before computing anything, the two objects must be compared as functions of their arguments.

	χ_D	$s_{\lambda/\mu}^c(x_1, \dots, x_\ell)$
a function of	the ordered tuple $D = (D_1, \dots, D_n)$, $D_j \subseteq [n]$	the pair $\mu \subseteq \lambda$ in $\mathcal{C}^{n,m}$ (equivalently $\lambda/d/\mu$) and ℓ
number of variables	$m_D := \max \bigcup_j D_j$, determined by D	ℓ , a free parameter independent of n, m
degree	$\#D = \sum_j \#D_j$, determined by D	$u(\lambda) - u(\mu)$, independent of ℓ
symmetric?	almost never — Theorem 4.5	always (Postnikov; reproved as cor:bk in the cylindric paper)
support	a sumset over columns (Lemma 2.1)	weights of chains of cylindric horizontal strips; adjacent steps are coupled

Remark 3.1 (A false obstruction, named and killed). The session brief conjectured an obstruction here: “*FMS’s variable count is tied to the grid; mine is a free parameter ℓ . If χ_D ’s variable count cannot be decoupled from the diagram, the two sides see different data and no direct application is possible.*” This is **false**, and it is worth killing explicitly rather than leaving as a background belief. If every column satisfies $D_j \subseteq [\ell]$ then $C \leq D$ forces $C_j \subseteq [\ell]$, so $\chi_D \in \mathbb{C}[x_1, \dots, x_\ell]$ — while the number of columns, and hence the degree, is bounded only by n , which may be taken as large as one likes by padding with empty columns. Concretely, $D_j = \{2\}$ for $j = 1, \dots, N$ (and empty thereafter) gives $\chi_D = h_N(x_1, x_2)$ for every N : two variables, unbounded degree. So the degree/variable ratio is no obstruction at all, and any argument resting on it would have been wrong.

Remark 3.2 (The cheap route is also blocked). There is a well-known soft route to results of the shape of Theorem 1.1: if f is Schur-positive, $\text{supp}(f) = \bigcup_\nu \text{supp}(s_\nu)$ over the ν occurring, and if that set of ν has a unique dominance maximum $\hat{\lambda}$ then SNP and $\text{Newton}(f) = \mathcal{P}_{\hat{\lambda}}$ follow from Rado’s theorem. This route is unavailable here: **cylindric skew Schur functions are not Schur-positive**. In my sample, 13 genuinely winding instances have a negative Schur coefficient, the smallest being $n = 3$, $m = 1$, $\mu = (0)$, $\lambda = (3)$, $\ell = 3$, where $s_{\lambda/\mu}^c = s_{21} - s_{111}$. (This is the same phenomenon as WZZ’s Example 4.2, $\tilde{F}_{s_2 s_1 s_0 s_2} = s_{22} - s_{1111}$, which is the published positive control in the cylindric paper’s §8.) Monomial positivity survives — s^c is a generating function over tableaux — but cancellation in the Schur basis is real, so the support cannot be read off a Schur expansion.

4 Symmetry rigidity

Throughout this section $D = (D_1, \dots, D_n)$ is a diagram, not all columns empty, and

$$m := \max \bigcup_j D_j.$$

Since $C_j \leq D_j$ implies $\max C_j \leq \max D_j$, Lemma 2.1 gives $\text{supp}(\chi_D) \subseteq \{\alpha \in \mathbb{N}^n : \alpha_i = 0 \text{ for } i > m\}$, and we regard $\text{supp}(\chi_D) \subseteq \mathbb{N}^m$. Every element has degree $N := \#D$, because $\#C_j = \#D_j$.

For $\alpha \in \mathbb{Z}^m$ write $\sigma_k(\alpha) = \alpha_1 + \dots + \alpha_k$ for its partial sums, and let $\alpha^\uparrow, \alpha^\downarrow$ denote the weakly increasing and weakly decreasing rearrangements. Let S_m act by permuting coordinates.

Definition 4.1. A column D_j is *bottom-justified* (in $[m]$) if $D_j = \emptyset$ or $D_j = \{m - \#D_j + 1, m - \#D_j + 2, \dots, m\}$.

4.1 Two distinguished points

Lemma 4.2. Put $\xi^\perp := \xi^D$ (take $C = D$) and $\xi^\top := \xi^C$ for $C_j = \{1, 2, \dots, \#D_j\}$. Then $\xi^\perp, \xi^\top \in \text{supp}(\chi_D)$ and, for every $\alpha \in \text{supp}(\chi_D)$ and every k ,

$$\sigma_k(\xi^\perp) \leq \sigma_k(\alpha) \leq \sigma_k(\xi^\top).$$

Explicitly $\xi_i^\perp = \#\{j : i \in D_j\}$ and $\sigma_k(\xi^\top) = \sum_j \min(\#D_j, k)$.

Proof. $C = D \leq D$ trivially. For the second, the t th least element of $\{1, \dots, r\}$ is t , and $t \leq d_t$ for any r -set $D_j = \{d_1 < \dots < d_r\}$, so $\{1, \dots, \#D_j\} \leq D_j$; both points lie in the support.

Let $C \leq D$ and fix k . Writing $C_j = \{c_1 < \dots < c_r\}$ and $D_j = \{d_1 < \dots < d_r\}$ with $c_t \leq d_t$ for all t ,

$$\#(C_j \cap [k]) = \#\{t : c_t \leq k\} \geq \#\{t : d_t \leq k\} = \#(D_j \cap [k]),$$

and also $\#(C_j \cap [k]) \leq \min(\#C_j, k) = \min(\#D_j, k) = \#(\{1, \dots, \#D_j\} \cap [k])$. Since $\sigma_k(\xi^C) = \sum_j \#(C_j \cap [k])$, summing over j gives both inequalities. $\square$

4.2 What symmetry forces

Lemma 4.3. Let $S \subseteq \mathbb{N}^m$ be finite, S_m -stable, with all elements of the same degree.

1. If $\mathbf{p} \in S$ satisfies $\sigma_k(\mathbf{p}) \leq \sigma_k(\alpha)$ for all $\alpha \in S$ and all k , then $\mathbf{p}$ is weakly increasing and $\text{sort}(\alpha) \trianglelefteq \text{sort}(\mathbf{p})$ for every $\alpha \in S$.
2. If $\mathbf{q} \in S$ satisfies $\sigma_k(\alpha) \leq \sigma_k(\mathbf{q})$ for all $\alpha \in S$ and all k , then $\mathbf{q}$ is weakly decreasing and $\text{sort}(\alpha) \trianglelefteq \text{sort}(\mathbf{q})$ for every $\alpha \in S$.

Proof. (1) Let s_i be the adjacent transposition. By stability $s_i \mathbf{p} \in S$, so $\sigma_i(s_i \mathbf{p}) \geq \sigma_i(\mathbf{p})$; but $\sigma_i(s_i \mathbf{p}) = \sigma_i(\mathbf{p}) - p_i + p_{i+1}$, whence $p_{i+1} \geq p_i$. So $\mathbf{p}$ is weakly increasing and $\sigma_k(\mathbf{p})$ is the sum of its k smallest entries.

Let $\alpha \in S$. By stability $\alpha^\uparrow \in S$, so $\sigma_k(\alpha^\uparrow) \geq \sigma_k(\mathbf{p})$ for all k ; that is, the sum of the k smallest entries of α is at least the sum of the k smallest entries of $\mathbf{p}$. Because α and $\mathbf{p}$ have the same total, this is equivalent to: the sum of the k largest entries of α is at most that of $\mathbf{p}$, for every k — i.e. $\text{sort}(\alpha) \trianglelefteq \text{sort}(\mathbf{p})$.

(2) is the mirror image, using $\sigma_i(s_i \mathbf{q}) \leq \sigma_i(\mathbf{q})$ and $\alpha^\downarrow \in S$. $\square$

Lemma 4.4. In the situation of Lemma 4.3, if both a $\mathbf{p}$ and a $\mathbf{q}$ exist then $\text{sort}(\mathbf{p}) = \text{sort}(\mathbf{q})$ and $\mathbf{p}_i = \mathbf{q}_{m+1-i}$ for all i .

Proof. $\mathbf{q} \in S$, so Lemma 4.3(1) gives $\text{sort}(\mathbf{q}) \trianglelefteq \text{sort}(\mathbf{p})$. $\mathbf{p} \in S$, so Lemma 4.3(2) gives $\text{sort}(\mathbf{p}) \trianglelefteq \text{sort}(\mathbf{q})$. Dominance is a partial order on partitions of a fixed size, so the two sorted vectors coincide. As $\mathbf{p}$ is weakly increasing and $\mathbf{q}$ weakly decreasing with the same multiset of entries, $\mathbf{p}$ is the reversal of $\mathbf{q}$. $\square$

4.3 The rigidity theorem

Theorem 4.5 (Symmetry rigidity). *Let D be a diagram with $m = \max \bigcup_j D_j$. Then $\text{supp}(\chi_D)$ is S_m -stable if and only if every column of D is bottom-justified in $[m]$.*

Proof. ($\Leftarrow$) If $D_j = \{m - k + 1, \dots, m\}$ with $k = \#D_j$, then an r -subset $\{a_1 < \dots < a_k\} \subseteq [n]$ is a basis of $\text{SM}_n(D_j)$ iff $a_t \leq m - k + t$ for all t . The condition at $t = k$ gives $a_k \leq m$, so the basis lies in $[m]$; and any k -subset of $[m]$ satisfies $a_t \leq m - k + t$ automatically, since at most $k - t$ of its elements exceed a_t . So the bases of $\text{SM}_n(D_j)$ are exactly the k -subsets of $[m]$, a manifestly S_m -stable family. By Lemma 2.1, $\text{supp}(\chi_D)$ is a sumset of S_m -stable sets, hence S_m -stable.

($\Rightarrow$) Suppose $\text{supp}(\chi_D)$ is S_m -stable. By Lemma 4.2 the hypotheses of Lemma 4.3 hold with $\mathbf{p} = \xi^\perp$ and $\mathbf{q} = \xi^\top$, so Lemma 4.4 gives $\xi_i^\perp = \xi_{m+1-i}^\top$ for all $i \in [m]$. Fix $k \in [m]$ and sum over the bottom k rows:

$$\sum_{i=m-k+1}^m \xi_i^\perp = \sum_{i=m-k+1}^m \xi_{m+1-i}^\top = \sum_{t=1}^k \xi_t^\top = \sigma_k(\xi^\top) = \sum_j \min(\#D_j, k),$$

the last equality by Lemma 4.2. On the other hand $\sum_{i=m-k+1}^m \xi_i^\perp = \sum_j \#(D_j \cap [m - k + 1, m])$. Since $\#(D_j \cap [m - k + 1, m]) \leq \min(\#D_j, k)$ term by term — the interval has k elements and D_j has $\#D_j$ — equality of the two sums forces equality for *each* j :

$$\#(D_j \cap [m - k + 1, m]) = \min(\#D_j, k) \quad \text{for all } j \text{ and all } k \in [m].$$

Now take $k = \#D_j$ (legitimate, as $\#D_j \leq m$): then $\#(D_j \cap [m - \#D_j + 1, m]) = \#D_j$, i.e. $D_j \subseteq [m - \#D_j + 1, m]$. That interval has exactly $\#D_j$ elements, so $D_j = \{m - \#D_j + 1, \dots, m\}$. $\square$

Remark 4.6. The proof uses symmetry only through the two adjacent-transposition inequalities and the two dominance comparisons of Lemma 4.3. In particular it never needs the full S_m -orbit structure of the support, and it is insensitive to the column *order* — as it must be, since $\text{supp}(\chi_D)$ is a sumset.

4.4 What the symmetric dual characters are

Corollary 4.7. *Suppose $\text{supp}(\chi_D)$ is S_m -stable, and let $\hat{\lambda}$ be the conjugate of the partition obtained by sorting the column sizes $(\#D_j)_j$ decreasingly. Then*

$$\text{supp}(\chi_D) = \{\alpha \in \mathbb{N}^m : \text{sort}(\alpha) \trianglelefteq \hat{\lambda}\} = \mathcal{P}_{\hat{\lambda}} \cap \mathbb{Z}^m = \text{supp}(s_{\hat{\lambda}}(x_1, \dots, x_m)),$$

and D is, up to reordering of columns, the skyline diagram of the antidominant rearrangement $(\hat{\lambda}_m, \hat{\lambda}_{m-1}, \dots, \hat{\lambda}_1)$, so that $\chi_D = \kappa_{(\hat{\lambda}_m, \dots, \hat{\lambda}_1)} = s_{\hat{\lambda}}(x_1, \dots, x_m)$.

Proof. By Theorem 4.5 each nonempty $D_j = \{m - k_j + 1, \dots, m\}$ with $k_j = \#D_j$, and (proof of Theorem 4.5, $\Leftarrow$) the bases of $\text{SM}_n(D_j)$ are all k_j -subsets of $[m]$; i.e. $\text{SM}_n(D_j)$ is the uniform matroid $U_{k_j, m}$ and $P(\text{SM}_n(D_j))$ is the hypersimplex $\Delta_{k_j, m} = \mathcal{P}_{(1^{k_j})}$. Minkowski sums of permutahedra add their partitions, so $\text{Newton}(\chi_D) = \sum_j \mathcal{P}_{(1^{k_j})} = \mathcal{P}_\nu$ with $\nu_i = \#\{j : k_j \geq i\} = \hat{\lambda}_i$. FMS Theorem 1.2 gives SNP, so $\text{supp}(\chi_D) = \mathcal{P}_{\hat{\lambda}} \cap \mathbb{Z}^m$, which is $\{\alpha : \text{sort}(\alpha) \trianglelefteq \hat{\lambda}\}$ by Rado's theorem.

For the last sentence: the skyline diagram of a composition α is $D(\alpha)_j = \{i : \alpha_i \geq j\}$ (§2.4(1)), which for α weakly increasing is the bottom-justified set of size $\#\{i : \alpha_i \geq j\}$. Taking α to be the antidominant rearrangement of $\hat{\lambda}$ reproduces the multiset $(k_j)_j$ of column sizes. FMS Theorem 5 (1.243, due to Reiner–Shimozono; see §2.4(2)) then identifies χ_D with κ_α , and $\kappa_\alpha = s_{\text{sort}(\alpha)}$ for α antidominant. Only the support statement is used below. $\square$

Corollary 4.8 (FMS Theorem 7 has no symmetric content beyond Rado). *Let D be a diagram whose dual character is symmetric in the variables it occupies. Then FMS Theorem 1.2 applied to D asserts exactly*

$$\text{supp}(s_{\hat{\lambda}}(x_1, \dots, x_m)) = \mathcal{P}_{\hat{\lambda}} \cap \mathbb{Z}^m, \quad \text{Newton}(s_{\hat{\lambda}}) = \mathcal{P}_{\hat{\lambda}},$$

for the $\hat{\lambda}$ of Corollary 4.7 — i.e. Rado’s theorem on the permutahedron (1952). The generality of FMS Theorem 7 is entirely in the non-symmetric directions.

5 Consequences for the cylindric theorem

5.1 The circularity

Theorem 5.1. *Let $\lambda/\mu \in \mathcal{C}^{n,m}$ and $\ell \geq 1$ with $\mathcal{T}_\ell(\lambda/\mu) \neq \emptyset$ and $u(\lambda) - u(\mu) \geq 1$. Suppose D is a diagram with*

$$\text{supp}(\chi_D) = \text{supp}(s_{\lambda/\mu}^c(x_1, \dots, x_\ell)).$$

Then $\max \bigcup_j D_j = \ell$; every column of D is bottom-justified in $[\ell]$; and the conjugate of the sorted column-size partition of D is exactly $\hat{\lambda}(\lambda/\mu, \ell)$. Equivalently: D is, up to column order, the skyline diagram of the antidominant rearrangement of $\hat{\lambda}$, and in particular D determines $\hat{\lambda}$ by inspection.

Proof. Write $W_\ell = \text{supp}(s_{\lambda/\mu}^c(x_1, \dots, x_\ell))$, which is S_ℓ -stable and consists of vectors of degree $N = u(\lambda) - u(\mu) \geq 1$. Since W_ℓ is S_ℓ -stable and nonempty with positive degree, some $\alpha \in W_\ell$ has $\alpha_\ell > 0$; as $\text{supp}(\chi_D) \subseteq \{\alpha : \alpha_i = 0 \text{ for } i > m_D\}$ with $m_D = \max \bigcup_j D_j$, we get $m_D \geq \ell$, and $m_D \leq \ell$ because $\xi_{m_D}^\perp > 0$. So $m_D = \ell$.

Now $\text{supp}(\chi_D) = W_\ell$ is S_ℓ -stable, so Theorem 4.5 applies: every column is bottom-justified. Corollary 4.7 gives $\text{supp}(\chi_D) = \{\alpha : \text{sort}(\alpha) \trianglelefteq \hat{\nu}\}$ for $\hat{\nu}$ the conjugate of the sorted column sizes. By Theorem 1.1, $W_\ell = \{\alpha : \text{sort}(\alpha) \trianglelefteq \hat{\lambda}\}$. Two dominance ideals of partitions of N with at most ℓ parts coincide iff their unique maxima coincide, so $\hat{\nu} = \hat{\lambda}$. $\square$

Theorem 5.1 is the answer to question (Q) of §1.2. A diagram D satisfying the hypothesis *does* exist for every $(\lambda/\mu, \ell)$ — take the skyline of the antidominant rearrangement of $\hat{\lambda}$ — so it is not true that FMS “cannot reach” the cylindric case. What is true is stronger and more precise:

Exhibiting a diagram D for which FMS Theorem 7 computes $\text{Newton}(s_{\lambda/\mu}^c(x_1, \dots, x_\ell))$ is the same act as computing $\hat{\lambda}(\lambda/\mu, \ell)$. The diagram is not an auxiliary object from which $\hat{\lambda}$ is later extracted; $\hat{\lambda}$ is literally its column-size partition, conjugated. There is no diagram attached to $(\lambda, \mu, d, n, m, \ell)$ by a construction that does not already contain the answer.

Corollary 5.2. *Theorem 1.1 is not a corollary of FMS Theorem 7. Any derivation of Theorem 1.1 from FMS Theorem 7 must supply, as an independent input, a partition $\hat{\lambda}$ and a proof that $\text{supp}(s_{\lambda/\mu}^c) = \{\alpha : \text{sort}(\alpha) \trianglelefteq \hat{\lambda}\}$ — which is Theorem 1.1 itself. Moreover, by Corollary 4.8, the FMS step in such a derivation would contribute only Rado’s theorem.*

5.2 The natural column dictionary fails at $d = 0$

The dictionary one would try first is: take the columns of the shape. For an ordinary skew shape λ/μ (the case $d = 0$, where $s_{\lambda/\mu}^c = s_{\lambda/\mu}$) these are $D_j = \{i : \mu_i < j \leq \lambda_i\}$, an honest diagram in FMS’s sense. It does not work, and Theorem 4.5 says exactly why: those columns are intervals determined by the shape’s own geometry, and they are bottom-justified only in degenerate cases.

Proposition 5.3. *Let λ/μ be an ordinary skew shape with r rows and let D be its column diagram in $[r]^2$. Then $\text{supp}(\chi_D) = \text{supp}(s_{\lambda/\mu}(x_1, \dots, x_r))$ only if every column of λ/μ ends in row r , i.e. only if λ/μ is (after deleting empty rows) a straight shape placed against the bottom of the grid.*

Proof. The right-hand side is S_r -stable, so by Theorem 4.5 every D_j is bottom-justified in $[m_D]$; and $m_D = r$ as in the proof of Theorem 5.1. A column of a skew shape is an interval $[a_j, b_j]$ of rows, and bottom-justified means $b_j = r$. Since μ is a partition, $b_j = r$ for all j forces $\mu_r = 0$ and $\mu_i \geq \lambda_{i+1}$ is vacuous; the shape is λ/μ with every column reaching the last row, which after deleting rows with no boxes is a straight shape. $\square$

The minimal witness is instructive and very small: $\lambda = (2, 1)$, $\mu = \emptyset$, $\ell = 2$, columns $D = (\{1, 2\}, \{1\})$. Here $\text{supp}(\chi_D) = \{(2, 1)\}$ — a single point — while $\text{supp}(s_{(2,1)}(x_1, x_2)) = \{(2, 1), (1, 2)\}$. The reason is transparent once seen: the straight shape $(2, 1)$ drawn at the top-left of the grid is the skyline diagram of the *dominant* composition $(2, 1)$, and $\kappa_{(2,1)} = x_1^2 x_2$. It is the *antidominant* placement that gives the Schur polynomial. A skew shape has no antidominant placement unless it is straight.

Remark 5.4 (The structural reason, in one sentence). $\text{supp}(\chi_D)$ is a sumset over columns (Lemma 2.1): the columns contribute independently, and the only constraint linking a filling to the diagram is the column-local flag $C_j \leq D_j$. A semistandard tableau, by contrast, couples *adjacent* columns through the weakly-increasing-along-rows condition. The flagged Weyl module replaces that coupling by a flag, and the flag is trivial precisely when the column is bottom-justified. That is Theorem 4.5 read backwards.

6 The complementary statement, and a correction to the brief

The session brief proposed, as the positive half of the question, the following target:

“The bead-hop exchange move of the cylindric paper is an explicit, finite certificate for the exchange axiom that FMS import non-constructively at l.190 via Schrijver Cor. 46.2c — restricted to the bases of $\text{SM}_\ell(D_j)$ for D_j a column of a cylindric skew shape.”

Having read the source, I record that **this statement is not correct and should not be proved**, on three counts.

1. FMS l.190 is the *definition* of a matroid, inside a background section. Nothing is imported there, and no exchange verification is needed anywhere in FMS: Schubert matroids are matroids by a cited classical fact (FMS l.194, citing *Oriented Matroids* §2.4).
2. What *is* imported non-constructively is a different statement at a different line: Schrijver, *Combinatorial Optimization* vol. B, Cor. 46.2c, at l.274 — the integer decomposition property for polymatroids, which is what yields SNP from the Minkowski sum description. The brief conflated a definition with an import.
3. The clause “for D_j a column of a cylindric skew shape” presupposes exactly the dictionary that Theorem 5.1 and Proposition 5.3 show does not exist. There is no diagram whose columns are the columns of a cylindric skew shape and whose dual character has the right support.

This is an instance of a pattern I have recorded before: a named obstruction — or here, a named opportunity — stated in a brief is never itself the object of a check. Both of the brief’s structural guesses (§3.1 on variable counts, and the passage above) are false, and the true obstruction is neither of them.

What survives is a genuine complementarity, which I state as a comparison of what the two proofs consume rather than as a new theorem:

Proposition 6.1 (Complementarity). *For a cylindric skew shape λ/μ and ℓ with $\mathcal{T}_\ell \neq \emptyset$, the M -convexity of $W_\ell = \text{supp}(s_{\lambda/\mu}^c(x_1, \dots, x_\ell))$ is proved in the cylindric paper by exhibiting, for each pair $\alpha, \beta \in W_\ell$ and each index i with $\alpha_i > \beta_i$, an explicit witness j and an explicit chain realising $\alpha - e_i + e_j \in W_\ell$: a single bead of one row of the tableau chain sliding one site to the left (**cor:exchange**). The route through FMS Theorem 7 instead obtains M -convexity as $\text{SNP} +$ (integral generalized permutahedron), where SNP rests on Schrijver Cor. 46.2c (FMS l.274) and the final step rests on Murota’s equivalence between M -convex sets and lattice point sets of integral generalized permutahedra — an equivalence FMS neither state nor cite, the word “ M -convex” not occurring in their paper. The two routes are therefore independent: neither uses an ingredient of the other.*

Proof. The first assertion is Corollary **cor:exchange** together with Proposition **prop:perm-mconvex** of the cylindric paper, whose “what the proof consumes” section (§9 there) lists Rado’s theorem, Lam Cor. 8.5, Lee 2019 Cor. 5, WZZ Thms 3.6 and 4.4, and Postnikov’s symmetry theorem as *not used*, and Murota’s theory as used only through the definition. The second assertion is a reading of FMS l.274 and of the absence of “ M -convex” from the source (grep, with brace-stripping, returns 0 occurrences in both **forArxiv.tex** and **forArxiv.bbl**). $\square$

7 Computational verification

All code is at **projects/proofs/code-q254/** and is committed. Every count below is reproducible by running **final.py**, **calibrate.py**, **rigidity.py**, **dictionary.py**, **winding.py**.

7.1 Is the transcription of FMS right?

The FMS side is implemented once, in **fms.py**, directly from the definitions quoted in §2: bases of $\text{SM}_n(D_j)$, then the sumset of their indicator vectors. It is calibrated against an implementation that shares no code and no concepts with it — Schubert and key polynomials computed from the divided-difference recursions of FMS l.78–96, with no matroids anywhere:

test	instances	agreement
$\text{supp}(\chi_{D(w)}) = \text{supp}(\mathfrak{S}_w)$, $w \in S_n$, $n \leq 5$	152	152/152
$\text{supp}(\chi_{D(\alpha)}) = \text{supp}(\kappa_\alpha)$, $\alpha \in \{0..n\}^n$, $n \leq 4$	698	698/698

This is a differential check on a definition transported from a paper into code, which is otherwise unfalsifiable inside the code. It also validates the corrected reading of the skyline formula (§2.4(1)): with FMS’s printed formula the key-polynomial test fails immediately.

7.2 Symmetry rigidity, exhaustive

rigidity.py enumerates *every* multiset of nonempty columns inside $[m]$ (column order is irrelevant, since the support is a sumset, so multisets are exhaustive), computes $\text{supp}(\chi_D)$, and tests S_m -stability against bottom-justification.

$m \leq$	$\# \text{cols} \leq$	diagrams	S_m -stable	sym. not b.j.	b.j. not sym.	sym. but $\neq \mathcal{P}_{\hat{\lambda}} \cap \mathbb{Z}^m$
4	5	15,503	205	0	0	0
5	4	52,359	246	0	0	0
6	3	45,759	203	0	0	0
		113,621	654	0	0	0

Both implications of Theorem 4.5 and the conclusion of Corollary 4.7 hold without exception on 113,621 diagrams.

7.3 Test A: the $d = 0$ calibration

`dictionary.py` compares $\text{supp}(\chi_D)$ for the natural column diagram of an ordinary skew shape against $\text{supp}(s_{\lambda/\mu}(x_1, \dots, x_\ell))$ computed by direct SSYT enumeration.

- 486 skew shapes with $|\lambda/\mu| \leq 8$ and at most 4 rows, $\ell =$ number of rows: the natural column diagram reproduces the true support in 166 cases and fails in **320**. In *exactly* the same 166 cases $\text{supp}(\chi_D)$ is S_ℓ -stable — perfect correlation with Theorem 4.5, and zero cases where the support is symmetric but wrong.
- Minimal witness: $\lambda = (2, 1)$, $\mu = \emptyset$, $\ell = 2$, $D = (\{1, 2\}, \{1\})$, $\text{supp}(\chi_D) = \{(2, 1)\}$ versus $\{(2, 1), (1, 2)\}$.
- Allowing the skew diagram to be placed at *any* vertical offset in an ℓ -row grid (537 triples $(\lambda/\mu, \ell)$, $|\lambda/\mu| \leq 6$, ≤ 3 rows, $\ell \leq 5$): some offset works in 261 cases and none works in 276. The ones that work are the straight shapes with room to be pushed to the bottom, as Proposition 5.3 predicts.

7.4 Test B: $d \geq 1$, and the uniqueness of the diagram

The sample is all $(\lambda/\mu, \ell)$ with $2 \leq n \leq 5$, $1 \leq m < n$, $|\lambda/\mu| \leq 6$, $2 \leq \ell \leq 4$: 488 instances.

Does the tested range actually wind? The cylinder enters the recursion at exactly one place, the wrap constraint $\rho_m < \nu_1 + n$. Replacing n by a large N with the same bead tuple relaxes it and computes the ordinary skew Schur function of the unrolled shape. An instance *winds* iff the two supports differ. Result: **199/488** instances wind. All the counts below are restricted to those 199.

Control on Theorem 1.1. All 199 winding instances have a unique dominance-maximal $\hat{\lambda}$ with $W_\ell = \{\alpha : \text{sort}(\alpha) \preceq \hat{\lambda}\}$. (This re-derives the cylindric paper’s headline on an independently generated sample; it is a control, not new evidence.)

Exhaustive search for a working diagram. For each distinct class $(\ell, N, \hat{\lambda})$ arising from a winding instance, `final.py` enumerates *every* diagram with columns inside $[\ell]$ and $\#D = N$, and keeps those with $\text{supp}(\chi_D) = W_\ell$. Result: in **36/36** classes there is **exactly one** solution, and it is the skyline diagram of the antidominant rearrangement of $\hat{\lambda}$. No other diagram works, at any placement, with any column multiset. This is Theorem 5.1 made concrete.

Negative controls.

- *Reversing the column order* was considered and **discarded as uninformative**: $\text{supp}(\chi_D)$ is a sumset over columns, so it is invariant under reordering by construction. Nothing about the world could make that control fail, so reporting it as a passing control would be reporting nothing. (It was run once as a check on the code path.)
- *Dropping one column* from the working diagram: 151/151 instances then fail to reproduce W_ℓ .
- *Moving one box of one column up by one row*, breaking bottom-justification: 158/158 instances then fail to reproduce W_ℓ .

The last two are controls that could have failed and did not.

Schur-positivity. Of the winding instances with $N \leq 5$, $\ell \leq 4$, 13 have a negative coefficient in the Schur basis; the smallest is $n = 3$, $m = 1$, $\mu = (0)$, $\lambda = (3)$, $\ell = 3$, where $s^c = s_{21} - s_{111}$. This is the computation behind Remark 3.2.

8 Verdict

FMS Theorem 7 does not reach the cylindric case, and the named obstruction is Theorem 4.5 (symmetry rigidity): the only diagrams whose dual character has symmetric support are the bottom-justified ones, whose dual characters are the straight-shape Schur polynomials. Consequently every diagram D with $\text{supp}(\chi_D) = \text{supp}(s_{\lambda/\mu}^c(x_1, \dots, x_\ell))$ has $\hat{\lambda}(\lambda/\mu, \ell)$ as the conjugate of its column-size partition, so producing D is producing $\hat{\lambda}$; and by Corollary 4.8 the FMS step then contributes exactly Rado’s theorem.

The registry node is **not** demoted on novelty. The obstruction has been promoted to a standalone proposition (Theorem 4.5), proved, and asked what would falsify it: a diagram with S_m -stable support and a column that is not bottom-justified. An exhaustive search over 113,621 diagrams found none, and the proof explains why none exists.

Grades: Theorem 4.5, Lemmas 4.2–4.4, Corollaries 4.7–4.8, Theorem 5.1, Proposition 5.3: **proved**. Proposition 6.1: **proved** as a statement about what the two arguments consume. All counts in §7: **computed**.

9 Gaps

1. **The verdict is about *direct* application of FMS Theorem 7 to $s_{\lambda/\mu}^c$.** It does not exclude an indirect argument in which Theorem 7 is one ingredient among several — for instance a decomposition $s_{\lambda/\mu}^c = \sum_D c_D \chi_D$ into dual characters followed by a support computation. I have not investigated such decompositions and do not claim they are impossible. I note only that by Remark 3.2 the coefficients c_D could not be assumed nonnegative, so no support statement would follow for free. Grade: this is an *open question*, not a claim.
2. **Corollary 4.7’s last sentence** (the identification $\chi_D = \kappa_\alpha = s_{\hat{\lambda}}$ as *polynomials*, not merely supports) cites FMS Theorem 5, i.e. Reiner–Shimozono JCTA 70 (1995), which I have not read. Nothing in §§5–7 uses it: every load-bearing statement is about supports, and the support version is proved here. Grade of the polynomial identification: **cited, unverified**.

3. **Rado's theorem** ($\mathcal{P}_{\hat{\lambda}} \cap \mathbb{Z}^m = \{\alpha : \text{sort}(\alpha) \trianglelefteq \hat{\lambda}\}$) is used in Corollary 4.7 and 4.8. It is classical, and the cylindric paper proves the instance it needs directly rather than citing it. Grade: `classical`, `used as stated`.
4. **FMS Theorem 10** (the Schubitope equals the same Minkowski sum) is a different statement that I have read but not separately analysed. Since it computes the same polytope $\sum_j P(\text{SM}_n(D_j))$, Theorem 4.5 applies to it verbatim; but I have not written that out. Grade: `not attempted`.
5. **The computational ranges are small** ($m \leq 6$ and at most 5 columns for the rigidity sweep; $|\lambda/\mu| \leq 6$, $\ell \leq 4$, $n \leq 5$ for the cylindric sweep). The rigidity theorem is proved, so its sweep is a check rather than evidence; the uniqueness sweep (§7, Test B) is likewise a corollary of the proved theorem. The only place where the range limits what is known is the $d = 0$ counts of Test A, which are descriptive statistics and are labelled as such.

Sources

- A. Fink, K. Mészáros, A. St. Dizier, *Schubert polynomials as integer point transforms of generalized permutahedra*, arXiv:1706.04935, Adv. Math. 332 (2018) 465–475. Read in full at first hand from the e-print LaTeX source (`projects/sources/1706.04935/forArxiv.tex`, 328 lines) on 2026-09-25; v1 and v2 byte-identical. Line numbers in §2 are my own.
- Clio, *Cylindric skew Schur functions have M-convex support* (`projects/proofs/2026-09-20-c1-cylindric-` Theorem `thm:main`, Corollary `cor:exchange`, §`sec:consumes`).
- A. Schrijver, *Combinatorial Optimization*, vol. B, Cor. 46.2c — cited only as the step FMS import at l.274; not used here.
- V. Reiner, M. Shimozono, *Key polynomials and a flagged Littlewood–Richardson rule*, JCTA 70 (1995) 107–143 — the true reference behind FMS's `\cite{keypolynomials}` (§2.4(2)); cited, not read.